EventHub

Event Ticket Booking System

A PHP and MySQL web application where people discover events, book tickets online, and administrators manage everything from one panel.

E-Service Course Project

ABOUT THE PROJECT

What is EventHub?

EventHub is a web-based E-Service for buying tickets to events such as concerts, seminars, workshops, tech meetups and sports matches. Visitors browse a catalogue, add tickets to a cart and complete a booking in a few steps. Organisers and administrators control the event listings, seat availability and every booking from a dedicated admin panel.

The problem

Event tickets are often sold through phone calls, chats or paper lists, which is slow and hard to track.

The solution

One website that shows live event information, available seats and a simple online booking flow.

The outcome

Fewer manual errors, instant booking records and a clear overview for the administrator.

GOALS

Project objectives

Easy discovery

Let users find events quickly by category, search and filters.

Simple booking

Cart, attendee details and payment choice in a short, clear flow.

Full admin control

Manage categories, events, posters, seats and bookings in one place.

Reliable data

Store users, events and bookings in a structured MySQL database.

Clean, valid code

Valid HTML5 and CSS with a simple, readable interface.

Customer updates

Email the customer when a booking is made, confirmed or cancelled.

USERS AND ROLES

Who uses the system

Visitor

- Browse events and categories
- Search and filter events
- View event details
- Register or log in

Registered user

- Everything a visitor can do
- Add tickets to the cart
- Book tickets and choose payment
- See own booking history

Administrator

- Manage categories and events
- Upload event posters
- Control available seats
- View, confirm or cancel bookings

FRONT-END

Features for customers

Event catalogue

Events listed with poster, title, date, venue and ticket price.

Categories

Music, Education, Sports, Tech and more, so users can narrow the list.

Register and login

Secure accounts with validated forms and a remember-me cookie.

Ticket cart

Add or remove tickets and change quantities before booking.

Booking checkout

Enter attendee details and pick a payment method (no real payment).

My bookings

Users can review their past and current bookings and their status.

BACK-END

Admin panel

Category management

Add, rename and delete event categories.

Event management

Create and edit events with title, description, date, venue, price and seats.

Poster upload

Upload an image for each event directly to the server.

Booking overview

See every booking with customer, tickets, total and status.

Booking details

Open a single booking to view all of its tickets and attendee data.

Seat control

Available seats go down when tickets are booked and return if cancelled.

GOING FURTHER

Advanced features

Search

Find events by name, description or price.

Smart filter

Combine category, city, date, language and price range in one query.

Event attributes

Flexible details such as city, language and age limit stored in related tables.

Confirm or cancel

Admin approves or cancels a booking and the status updates instantly.

Email notifications

Automatic emails after booking and after confirmation or cancellation.

WORKFLOW

How a booking works

1. **Browse** — User explores events and categories
2. **Select** — Opens an event and picks ticket quantity
3. **Cart** — Tickets are held in the session cart
4. **Checkout** — Enters attendee details and payment method
5. **Confirm** — Booking saved, seats reduced, email sent
6. **Review** — Admin confirms or cancels the booking

Booking status moves from **Pending** to **Confirmed** or **Cancelled**. A cancelled booking returns its seats to the event.

MYSQL

Database design

users

id (PK)
name
email
password_hash
role

orders

id (PK)
user_id (FK)
attendee_name
payment_method
status
created_at

order_items

id (PK)
order_id (FK)
event_id (FK)
quantity
price

events

id (PK)
category_id (FK)
title
description
poster
event_date
venue
price
seats_available

categories

id (PK)
name

attributes

id (PK)
name (city, language...)

event_attributes

event_id (FK)
attribute_id (FK)
value

Relations: users 1-N orders | orders 1-N order_items | events 1-N order_items | categories 1-N events | events N-N attributes (via event_attributes)

STACK

Technology and design

PHP

Server-side logic: forms, sessions, cookies, file upload and the mail() function.

MySQL

Relational database with joins for filtering and reporting.

HTML5 and CSS3

Valid, semantic markup checked with the W3C validator.

Bootstrap

Clean, responsive layout that works on phones and desktops.

QUALITY

Validation and security

Input validation

All form data is checked on the server before it is used.

Password hashing

Passwords are stored with password_hash(), never as plain text.

SQL injection protection

Prepared statements for every database query.

Access control

Admin pages are only reachable by users with the admin role.

Safe uploads

Only image files of limited size are accepted for posters.

Session safety

Sessions are regenerated at login and cleared at logout.

WRAP-UP

Summary

What it delivers

A complete ticket booking service: catalogue, accounts, cart, booking and admin control.

What it demonstrates

PHP, MySQL, forms, sessions and cookies, file upload, SQL joins and email.

Why it works

A realistic use case with a clear flow, easy to explain and to extend later.

Possible future additions: online payment gateway, QR-code tickets, seat maps and event reviews.